Notation:

Let $\mathbb{C}_L^+ = \{z : \mathrm{im}(z) > L\}$ and $B_r = \{f \in H^\infty(\mathbb{C}_L^+) : \|f\| \le r\}$

Denote the map $T_\mu(f)(z) = R_\mu\left(\frac{1}{z - f(z)}\right)$. Then we have the following theorem.

Theorem 1. Given compactly supported probability measure μ, with $\mathrm{supp}(\mu) \subset [-K, K]$. Then, for any lipstchiz constant $q > 0$ and containment constant $d \ge 0$, we can find a suitable region B_r such that

1. $T_\mu(B_{dr}) \subset B_r$
2. $\left\|T_\mu(f) - T_\mu(g)\right\|_\infty \le q\|f - g\|_\infty$

Proof. Let $M = \sup_{|z| \le \frac{1}{6K}} \left|R'_\mu(z)\right| < \infty$.

Choose $r > R_\mu(0)$ and sufficient large L such that $\frac{1}{L - dr} < \frac{1}{6K}$, $\frac{M}{L - dr} < r - \left|R_\mu(0)\right|$, $\frac{M}{(L-dr)^2} < q$

- $T_\mu(B_{dr}) \subset B_r$

According to [1], for compactly supported measure R_μ is analytic on $\left\{z : |z| \le \frac{1}{6K}\right\}$. For $f \in B_{dr}$, we have $|f(z)| \le dr$ for $z \in \mathbb{C}_L^+$. It follows that $|z - f(z)| \ge |z| - |f(z)| \ge L - dr \ge 6K$, thus $T_\mu f$ is well defined on B_{dr}.

Next, we need to show $\left\|T_\mu f\right\|_\infty \le r$. To see this, observe that for $|z| \le \frac{1}{L - dr}$, we have

$$\begin{aligned}\left|R_\mu(z)\right| \le \left|R_\mu(0)\right| + \left|R_\mu(z) - R_\mu(0)\right| &\le \left|R_\mu(0)\right| + |z| \sup_{w \in [0,z]} \left|R'_\mu(w)\right| \\ &\le \left|R_\mu(0)\right| + \frac{M}{L - dr} \le r\end{aligned}$$

Taking supremum on both side yields $\left\|T_\mu f\right\|_\infty \le r$

- $\left\|T_\mu(f) - T_\mu(g)\right\|_\infty \le q\|f - g\|_\infty$

Notice that for $z \in \mathbb{C}_L^+$

$$\begin{aligned}\left|T_\mu f(z) - T_\mu g(z)\right| = \left|R_\mu\left(\frac{1}{z - f(z)}\right) - R_\mu\left(\frac{1}{z - g(z)}\right)\right| &\le \frac{M}{(L - dr)^2}|f(z) - g(z)| \\ &\le q|f(z) - g(z)|\end{aligned}$$

Again, taking supremum yields $\left\|T_\mu f - T_\mu g\right\| \le q\|f - g\|$ □

Remark 2. It is well known that for free and boolean cumulant-cumulant relations. We have the following conversion.

$$\beta_m = \sum_{\pi \in \mathrm{NC}_{\mathrm{irr}}(m)} \kappa_\pi$$

where $\mathrm{NC}_{\mathrm{irr}}$ mens that $1, m$ are in the same blocks.

Now, since we have $R_\mu(z) = \sum_{i \ge 0} \kappa_{i+1} z^i$, then the (formally) expansion of the iterated operator is

$$T_\mu f(z) = R_\mu\left(\frac{1}{z - f(z)}\right) = \sum_{i \ge 0} \kappa_{i+1}(z - f(z))^{-i} = \sum_{i \ge 0} \kappa_{i+1} z^{-i} \sum_{j \ge 0} \binom{i + j - 1}{j} \left(\frac{f(z)}{z}\right)^j$$

Take $f_0 = 0$ and $f_n = T_\mu f_{n-1}$, a first few terms of this sequence is

$$
\begin{aligned}
&f_0(z) = 0\\
&f_1(z) = \kappa_1 + \kappa_2 z^{-1} + \kappa_3 z^{-2} + \kappa_4 z^{-3} + \kappa_5 z^{-4} + ...\\
&f_2(z) = \kappa_1 + \kappa_2 z^{-1} + (\kappa_3 + \kappa_1\kappa_2) z^{-2} + (\kappa_4 + 2\kappa_1\kappa_3 + \kappa_2^2 + \kappa_1^2\kappa_2) z^{-3} +\\
&(\kappa_5 + 3\kappa_2\kappa_3 + 3\kappa_1\kappa_4 + 3\kappa_1^2\kappa_3 + 2\kappa_1\kappa_2^2 + \kappa_1^3\kappa_2) z^{-4} + ...\\
&f_3(z) = \kappa_1 + \kappa_2 z^{-1} + (\kappa_3 + \kappa_1\kappa_2) z^{-2} + (\kappa_4 + 2\kappa_1\kappa_3 + \kappa_2^2 + \kappa_1^2\kappa_2) z^{-3} +\\
&(\kappa_5 + 3\kappa_2\kappa_3 + 3\kappa_1\kappa_4 + 3\kappa_1^2\kappa_3 + 3\kappa_1\kappa_2^2 + \kappa_1^3\kappa_2) z^{-4} + ...\\
&f_4(z) = \kappa_1 + \kappa_2 z^{-1} + (\kappa_3 + \kappa_1\kappa_2) z^{-2} + (\kappa_4 + 2\kappa_1\kappa_3 + \kappa_2^2 + \kappa_1^2\kappa_2) z^{-3} +\\
&(\kappa_5 + 3\kappa_2\kappa_3 + 3\kappa_1\kappa_4 + 3\kappa_1^2\kappa_3 + 3\kappa_1\kappa_2^2 + \kappa_1^3\kappa_2) z^{-4} + ...
\end{aligned}
$$

Now, lets see the irreducible non-crossing partitions with boolean cumulant.

$$
\begin{aligned}
\beta_1 &= \kappa_1\\
\beta_2 &= \kappa_2\\
\beta_3 &= \kappa_3 + \kappa_1\kappa_2\\
\beta_4 &= \kappa_4 + 2\kappa_1\kappa_3 + \kappa_2^2 + \kappa_1^2\kappa_2\\
\beta_5 &= \kappa_5 + 3\kappa_2\kappa_3 + 3\kappa_1\kappa_4 + 3\kappa_1^2\kappa_3 + 3\kappa_1\kappa_2^2 + \kappa_1^3\kappa_2
\end{aligned}
$$

Thus, we have discover that the relation is

$$
\begin{aligned}
&f_0(z) = 0\\
&f_1(z) = \beta_1 + \beta_2 z^{-1} + \kappa_3 z^{-2} + \kappa_4 z^{-3} + \kappa_5 z^{-4} + ...\\
&f_2(z) = \beta_1 + \beta_2 z^{-1} + \beta_3 z^{-2} + \beta_4 z^{-3} + (\kappa_5 + 3\kappa_2\kappa_3 + 3\kappa_1\kappa_4 + 3\kappa_1^2\kappa_3 + 2\kappa_1\kappa_2^2 + \kappa_1^3\kappa_2) z^{-4} + ...\\
&f_3(z) = \beta_1 + \beta_2 z^{-1} + \beta_3 z^{-2} + \beta_4 z^{-3} + \beta_5 z^{-4} + ...\\
&f_4(z) = \beta_1 + \beta_2 z^{-1} + \beta_3 z^{-2} + \beta_4 z^{-3} + \beta_5 z^{-4} + ...
\end{aligned}
$$

This iteration has a nice combinatorial interpretation. The n-th iteration of T_μ gives the first $2n$ boolean cumulants, and the rest are still free cumulants.

Remark 3. In particular, take $k > 1, q = 1$, we have T_μ be a contraction. In this case, according to Banach fixed point theorem, we have a unique fixed point $B_\mu(z) = z - \frac{1}{G_\mu(z)}$.

Proof. We will show that $B_\mu(z) \in B_r \subset B_{dr}$. To see this, for $z \in \mathbb{C}_L^+$ given above, we have

$$
\left|G_\mu(z)\right| \leq \int \left|\frac{1}{z-t}\right| d\mu \leq \int \frac{1}{\operatorname{im}(z)} d\mu = \frac{1}{\operatorname{im}(z)} \leq \frac{1}{L} \leq \frac{1}{L - dr}
$$

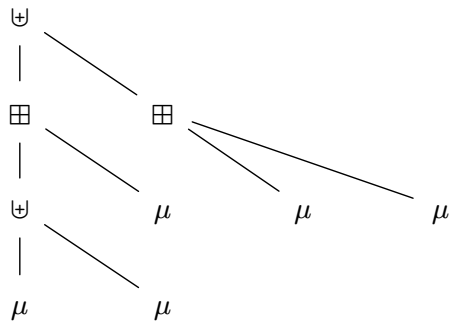
Thus, $\left|B_\mu(z)\right| = \left|R_\mu\big(G_\mu(z)\big)\right| \leq r$ from the above arguments, we then have $\left\|B_\mu(z)\right\|_\infty \leq r$. Last, to show it is a fixed point. Consider $T_\mu\big(B_\mu(z)\big) = R_\mu\Big(\frac{1}{z - B_\mu(z)}\Big) = R_\mu\big(G_\mu(z)\big) = B_\mu(z)$. □

Remark 4. Next, consider the product space $H_L^N \coloneqq H^\infty(\mathbb{C}_L^+)^N$ and the subspace $X \coloneqq B_{dr}^N \subset H_L^N$. Define the linear map $D \in B\big(H_L^N\big)$ by $D = D_B \otimes I$ for some $D_B \in M_d(\mathbb{C})$ and let $d = \|D\|_\infty < \infty$.

Consider the map $\boldsymbol{T}_\mu : X \to B_r^N$ by $(f_1, ...f_N) \mapsto \big(T_\mu f_1, ...T_\mu f_N\big)$. Then, we are interested in $\Phi = D \circ \boldsymbol{T}_\mu$. First, note that since $\|D\|_\infty = d$, we have $D : B_r^N \to B_{dr}^N = X$. Thus, $\Phi : X \to X$. Also, since $\|\Phi(F) - \Phi(G)\|_\infty \leq \|D\|_\infty \big\|\boldsymbol{T}_\mu F - \boldsymbol{T}_\mu G\big\| \leq d\frac{M}{(L-dr)^2}\|F - G\|_\infty$. Then, by choosing sufficiently large L, we have Φ is also a contraction.

Definition 5. We will called $\mathcal{T}$ an expression tree for any arithmetic combinations if the leaf node of this tree is number and the internal node is operator. Note that we will use t_{ij} to denote the ith-level jth-node starting from top left. Also, the leaf node is denoted by $t_{\text{leaf},j}$.

Example 1. For the case $\nu := ((\mu \uplus \mu) \boxplus \mu) \uplus (\mu \boxplus \mu)$, the corresponding expression tree is



Remark 6. Every arithmetic operations can be expressed by an expression tree.

Theorem 7. Let μ be a compactly supported probability measure with $\operatorname{supp}(\mu) \subset [-K, K]$, and let $\mathcal{T}_\nu$ be the expression tree of ν which is obtained from mixing free and boolena convolutions. Then, $\mathcal{T}_\nu$ admits a fixed point equations set $X = AX + E_B T_\mu(C_B X)$.

Proof.

Let $\mu_{i,j}$ denote the measure represented by the subtree rooted at $t_{i,j}$. That is, for every internal node $t_{i,j}$, we have $\mu_{i,j} = \mu_{i+1,2j} t_{i,j} \mu_{i+1,2j+1}$, where $t_{i,j} \in \{\uplus, \boxplus\}$ and for every leaf node $t_{\text{leaf},k}$, we have $\mu_{\text{leaf},k} = \mu$, For each node, we introduce a state variable $x_{i,j}$ and a linear form $l_{i,j}(X)$ such that $x_{i,j} = B_{\mu_{i,j}}\big(z - l_{i,j}(X)\big)$.

At the root, we set $l_{0,0}(X) = 0$, and therefore $x_{0,0} = B_\nu(z)$. We define the $l_{i,j}(X)$ and the child states as follows:

If $t_{i,j} = \uplus$, then $B_{\mu_{i,j}} = B_{\mu_{i+1,2j}} + B_{\mu_{i+1,2j+1}}$. Hence we set

$$l_{i+1,2j}(X) = l_{i+1,2j+1}(X) = l_{i,j}(X),$$

It follows that

$$x_{i,j} = x_{i+1,2j} + x_{i+1,2j+1}.$$

If $t_{i,j} = \boxplus$, then $x_{i,j} = R_{\mu_{i+1,2j}}\left(\frac{1}{z - l_{i,j}(X) - x_{i,j}}\right) + R_{\mu_{i+1,2j+1}}\left(\frac{1}{z - l_{i,j}(X) - x_{i,j}}\right)$. Define

$$l_{i+1,2j}(X) = l_{i,j}(X) + x_{i,j} - x_{i+1,2j}$$
$$l_{i+1,2j+1}(X) = l_{i,j}(X) + x_{i,j} - x_{i+1,2j+1}$$

Then, using the uniqueness of the fixed point, we obtain $x_{i+1,2j} = B_{\mu_{i+1,2j}}\big(z - l_{i,j}(X) - x_{i,j} + x_{i+1,2j}\big) = R_{\mu_{i+1,2j}}\left(\frac{1}{z - l_{i,j}(X) - x_{i,j}}\right)$ and similarly for $x_{i+1,2j+1}$ Then,

$$x_{i,j} = x_{i+1,2j} + x_{i+1,2j+1}.$$

Therefore, for every internal node,

$$x_{i,j} = x_{i+1,2j} + x_{i+1,2j+1},$$

while the argument shifts satisfy the recursive rule

$$l_{\text{child}}(X) = \begin{cases} l_{\text{parent}}(X) \text{ if the parent is a Boolean node} \\ l_{\text{parent}}(X) + x_{\text{parent}} - x_{\text{child}} \text{ if the parent is a free node.} \end{cases}$$

Since the tree is finite, this procedure eventually reaches every leaf. For the k-th leaf,

$$x_{\text{leaf},k} = B_\mu\big(z - l_{\text{leaf},k}(X)\big).$$

By the fixed-point identity of B_μ,

$$x_{\text{leaf},k} = T_\mu\big(l_{\text{leaf},k}(X) + x_{\text{leaf},k}\big)$$

Define the complete input of the k-th leaf by

$$L_k(X) \coloneqq l_{\text{leaf},k}(X) + x_{\text{leaf},k}.$$

Now let $X = (x_t)_{t \in \mathcal{T}_\nu}$ be the vector containing the states of all nodes of the tree.

Let $A = \big(a_{i,j}\big)$ be the directed parent-to-children adjacency matrix, that is

$$a_{i,j} = \begin{cases} 1 \text{ if node } i \text{ is a child of the internal node } j \\ 0 \text{ otherwise.} \end{cases}$$

Next, define C_B by

$$C_B X = \begin{pmatrix} L_1(X) \\ L_2(X) \\ \vdots \\ L_N(X) \end{pmatrix}$$

Let

$$\boldsymbol{T}_\mu \begin{pmatrix} f_1 \\ \vdots \\ f_N \end{pmatrix} = \begin{pmatrix} T_\mu(f_1) \\ \vdots \\ T_\mu(f_N) \end{pmatrix}.$$

Finally, let E_B be the leaf embedding matrix which places the N nonlinear outputs into the corresponding leaf coordinates.

Therefore,

$$X = AX + E_B \boldsymbol{T}_\mu(C_B X).$$

□

Remark 8. This fixed point equation is equivalent to

$$Y = C_B(I - A)^{-1} E_B \boldsymbol{T}_\mu(Y) = D_B \boldsymbol{T}_\mu(Y) = \Phi(Y)$$

Note that according to Theorem 1, Φ is a contraction map.

Example 2. For $\nu \coloneqq ((\mu \boxplus \mu) \uplus \mu) \boxplus \mu$, we have 7 states and the following fixed point equations

$$\begin{cases} x_0 = B_\nu = x_1 + x_2 \\ x_1 = x_3 + x_4 \\ x_2 = T_\mu(B_\nu) = T_\mu(x_0) \\ x_3 = x_5 + x_6 \\ x_4 = B_\mu(z - x_0 + x_1) = T_\mu(x_0 - x_1 + x_4) \\ x_5 = B_\mu(z - x_0 + x_1 - x_3 + x_5) = T_\mu(x_0 - x_1 + x_3) \\ x_6 = B_\mu(z - x_0 + x_1 - x_3 + x_6) = T_\mu(x_0 - x_1 + x_3) \end{cases}$$

reordering the leaves node in to the last, through matrix notation, we have

$$X = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} X + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} T_\mu \left(\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix} X \right)$$

Then, the corresponding reduced fixed point equation is

$$Y = \Phi(Y) = D_B \boldsymbol{T}_\mu(Y) = C_B(I - A)^{-1} E_B \boldsymbol{T}_\mu(Y) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{pmatrix} \boldsymbol{T}_\mu(Y)$$

Also, in this case, we have $\|D_B\|_\infty = 4$, thus, according to Remark 4, we can choose sufficiently large L such that Φ is a contraction.

Remark 9. However, lets look at Example 2 again, we can actually do further reduction. Consider the following equations

$$\begin{cases} y_0 = T_\mu(y_0) + T_\mu(y_1) + T_\mu(y_2) + T_\mu(y_3) \\ y_1 = T_\mu(y_0) + T_\mu(y_1) \\ y_2 = T_\mu(y_0) + T_\mu(y_2) + T_\mu(y_3) \\ y_3 = T_\mu(y_0) + T_\mu(y_2) + T_\mu(y_3) \end{cases}$$

It is easy to see that $y_2 = y_3$, thus we can reduce the equations to

$$\begin{cases} y_0 = T_\mu(y_0) + T_\mu(y_1) + 2T_\mu(y_2) \\ y_1 = T_\mu(y_0) + T_\mu(y_1) \\ y_2 = y_0 - T_\mu(y_1) \end{cases}$$

Note that in the second equation, we have $y_1 = T_\mu(y_0) + T_\mu(y_1)$, then by observing the identity,

$$y_1 - T_\mu(y_0) = T_\mu(y_1) = R_\mu \left(\frac{1}{z - T_\mu(y_0) - \big(y_1 - T_\mu(y_0)\big)} \right)$$

and using the uniqueness of the fixed point, we have $y_1 - T_\mu(y_0) = B_\mu\big(z - T_\mu(y_0)\big)$, thus we have the following reduced equations

$$y_0 = T_\mu(y_0) + T_\mu\big(T_\mu(y_0) + B_\mu\big(z - T_\mu(y_0)\big)\big) + 2T_\mu\big(y_0 - T_\mu\big(T_\mu(y_0) + B_\mu\big(z - T_\mu(y_0)\big)\big)\big)$$

Since $y_0 = B_\nu(z)$, we have reduced the equation set into 1 dimension.

Remark 10. Not all the fixed point equations set can be reduced to 1 dimension using the method of substitution(or can we?). For example, consider $\nu = ((\mu \uplus \mu) \boxplus \mu) \uplus (\mu \boxplus \mu)$, we have the following

Since we know that $B_{\mu\boxplus\mu}(z) = 2B_\mu\big(z - \frac{1}{2}B_{\mu\boxplus\mu}(z)\big)$, we have

$$R_\nu(z) = B_{(\mu\uplus\mu)\boxplus\mu}\left(R_\nu(z) + \frac{1}{z}\right) + B_{\mu\boxplus\mu}\left(R_\nu(z) + \frac{1}{z}\right)$$

$$B_{(\mu\uplus\mu)\boxplus\mu}(z) = R_{\mu\uplus\mu}\left(\frac{1}{z - B_{(\mu\uplus\mu)\boxplus\mu}(z)}\right) + R_\mu\left(\frac{1}{z - B_{(\mu\uplus\mu)\boxplus\mu}(z)}\right)$$

$$R_{\mu\uplus\mu}(z)) = 2B_\mu\left(R_{\mu\uplus\mu}(z) + \frac{1}{z}\right)$$

Substituting this all together yields

$$\begin{cases} R_\nu & = B_{\mu\boxplus\mu}\big(R_\nu(z) + \frac{1}{z}\big) + R_\mu\left(\frac{1}{\frac{1}{z} + B_{\mu\boxplus\mu}(R_\nu(z) + \frac{1}{z})}\right) \\ & \quad + 2B_\mu\left(R_\nu(z) + \frac{1}{z} - R_\mu\left(\frac{1}{\frac{1}{z} + B_{\mu\boxplus\mu}(R_\nu(z) + \frac{1}{z})}\right)\right) \\ B_{\mu\boxplus\mu}\big(R_\nu(z) + \frac{1}{z}\big) & = 2B_\mu\big(R_\nu(z) + \frac{1}{z} - \frac{1}{2}B_{\mu\boxplus\mu}\big(R_\nu(z) + \frac{1}{z}\big)\big) \end{cases}$$

or simplifying with $h = B_{\mu\boxplus\mu}\big(R_\nu(z) + \frac{1}{z}\big)$, we can get the following equations set.

$$\begin{cases} R_\nu(z) = h(z) + R_\mu\left(\frac{1}{\frac{1}{z} + h(z)}\right) + 2S_\mu\left(R_\nu(z) + \frac{1}{z} - R_\mu\left(\frac{1}{\frac{1}{z} + h(z)}\right)\right) \\ h(z) \quad = 2S_\mu\big(R_{\nu(z)} - \frac{1}{2}h(z)\big) \end{cases}$$

Some questions to think about

- What is the conditions to reduce the fixed point equations set into 1 dimension?
- Can we extend the strategy into more general case, for example, for probability measure.
- Can we characterize the convergence theorem of mixing convolutions?
- Can we say something about the asymptotic behavior like $\left(\mu^{\boxplus M}\right)^{\uplus N}$?
- What are the tree invariants for this fixed point equations?

At this point, we will study the asymptotic behavior of the fixed point equations that can be reduced to one dimensional fixed point equation. In particular, we will starts from the basic CLT theorem and then for the known results of $\left(\mu^{\boxplus M}\right)^{\uplus N}$ and the for $\mu \boxplus \mu \uplus \mu \boxplus \mu \ldots$. At the end, we will give a algorithm to compute the asymptotic distribution of the fixed point equations set that can be reduced to 1 dimension.

Example 3. For free CLT, for any compactly supported probability measure ν with centered mean and variance 1, let $\nu_n = D_{\frac{1}{\sqrt{n}}}\nu$ Then, for $\nu_n^{\boxplus n}$, we have the following fixed point equation

$$\begin{cases} x_0 = x_1 + x_2 \\ x_2 = T_{\nu_n}(x_0) \\ x_1 = x_3 + x_4 \\ \vdots \end{cases}$$

Inductively, we have the reduced one dimensional fixed point equations $x_0 = nT_{\nu_n}(x_0) = nR_{\nu_n}\big(\frac{1}{z} - x_0\big) = \sqrt{n}R_\nu\Big(\frac{1}{\sqrt{n}(z - x_0)}\Big)$. Denote $\Phi_n(f) = \sqrt{n}R_\nu\Big(\frac{1}{\sqrt{n}(z-f)}\Big)$

One can show that $\Phi_n(f) \to \Phi_\infty(f) = \frac{1}{z-f}$. Then, for any fixed point satisfy $\Phi_\infty(f_\infty) = f_\infty$, we have $f_\infty = \frac{z-\sqrt{z^2-4}}{2}$, which is the Cauchy (Boolean) transform of the semicircle distribution.

Example 4. Next, we will consider the case of $\left(\mu^{\boxplus M}\right)^{\uplus N}$. In this case, we have the following fixed point equations set

Bibliography

[1] R. S. James A. Mingo, *Free Probability and Random Matrices*. Springer, 2017.